Band-1 Computational Experiment: Exact Capset Sizes a(n) for Small n

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Band-1 Computational Experiment: Exact Capset Sizes a(n) in (Z/3Z)^n

Honesty stamp. This note does **not** claim that the asymptotic cap-set problem is solved. Ellenberg–Gijswijt (2017) is an **asymptotic** upper bound $a(n) = o(2.756^n)$. The Band-1 target here is **exact small-n** verification and a frontier ledger for the next open exact n.

Abstract

We probe the classical cap-set function $a(n)$: the maximum size of a subset of $(\mathbb{Z}/3\mathbb{Z})^n$ with no three-term affine line. On this Linux box we **fully certify** $a(n)$ for $n \leq 3$ by exhaustive enumeration, and we **certify** $a(4) \geq 20$ by an explicit verified witness. Literature: $a(4) = 20$, $a(5) = 45$, $a(6) = 112$. **Next open exact n: 7** ($a(7) \geq 236$).

Problem statement

Three distinct points $x, y, z \in \mathbb{F}_3^n$ form an affine line iff $x + y + z = 0$. A capset has no such triple. OEIS A090245 begins $a(0), \ldots, a(6) = 1, 2, 4, 9, 20, 45, 112$, with $a(7)$ open.

Method

Enumeration ($n \leq 3$). Lexicographic DFS: take point p, forbid thirds $-(p + s)$ for chosen s. Then $a(n) = \max |C|$.

Branch-and-bound. Bitset DFS with symmetry break (force $\{0, 1\} \subset C$ for $n \geq 2$).

Slice construction ($n = 4$). Decompose by last coordinate into slices C_0, C_1, C_2 in $\mathbb{F}_3^3$; enforce multiplicity ≤ 2 and the cross-slice rule $x \in C_0, y \in C_1 \Rightarrow -(x + y) \notin C_2$. Search yields a 20-point witness.

Results (numbers!)

Box-certified exact values

n	$a(n)$	Capsets enumerated	Enum (s)	BB (s)	Match lit.
0	**1**	2	2.3×10^{-5}	—	yes

n	$a(n)$	Capsets enumerated	Enum (s)	BB (s)	Match lit.
1	**2**	7	2.9×10^{-5}	2.5×10^{-5}	yes
2	**4**	172	2.3×10^{-4}	2.8×10^{-5}	yes
3	**9**	**367525**	**1.161**	3.1×10^{-4}	yes

Dim-3 sizes 0..9 counts: 1, 27, 351, 2808, 14742, 50544, 107406, 126360, 63180, 2106.

$n = 4$ **lower bound**

- Witness size **20**, `is_capset=true`, search ≈ 0.034 s.
- **Box-certified:** $a(4) \geq 20$.
- Literature: $a(4) = 20$ (Pellegrino 1971). Partial slice-pair upper census on-box: completed pairs (including $(9,9)$, $(9,8)$, $(8,8)$) had local max ≤ 20; no exceedance. Full pair census not finished in this wrap.

Literature and frontier

n	Status
4	$= 20$ lit; ≥ 20 box
5	$= 45$ lit
6	$= 112$ lit
7	**open**; ≥ 236
8	**open**; ≥ 512

Next open exact n: 7.

Asymptotics (not claimed solved)

Ellenberg–Gijswijt (2017): $a(n) = o(2.756^n)$. This does not determine $a(7)$.

Reproducibility

- Code: `/workspace/experiments/band1/capset_exact/`
- Results: `results.json`
- Log: `/workspace/experiments/logs/2026-09-11_capset_exact.log`

Conclusion

$a(0..3) = (1, 2, 4, 9)$ **box-certified**; $a(4) \geq 20$ **box-certified**; literature $a(4..6) = (20, 45, 112)$; **frontier = exact** $a(7)$. No asymptotic overclaim.